

Azure Database for PostgreSQL Flexible Server Resiliency Architecture

WHITEPAPER V2.0

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Introduction

Azure Database for PostgreSQL flexible server provides robust protection against regional and zonal failures through an array of advanced features. This white paper examines these features, illustrating how they can be utilized to build a resilient architecture for flexible server deployments. By comprehending and implementing these capabilities, you can ensure optimal uptime and reliability for your mission-critical applications.

Achieving Resiliency

When you deploy your database/application in the public cloud, it is important to look beyond the VM that is hosting your application or the Flexible Server that is running your database. Azure, like any public or private data center, consists of a set of services collaborating to create a resilient environment to run your workload. In this context, it is important to revisit what failures can happen and how Azure provides resiliency through those failures.

Zonal Resiliency: Ability of a database to withstand failures within a specific availability zone. Deploying primary and standby instances in different availability zones can help achieve zonal resiliency.

Regional Resiliency: Ability of a system to withstand failures that affect an entire region. Creating read-replicas, enabling geo-redundant backups helps in achieving regional resiliency for your database.

Infrastructure Outages and resiliency

Infrastructure consists of data centers with compute, storage, memory, power-supply, and networks as described below

- **Hardware failure:** Failures in the underlying hardware components such as CPUs, memory, disks, or network interfaces
Azure provides hardware resiliency by automatically creating the failed virtual machine
- **Data Center outages:** Entire data center outages due to power failures, natural disasters, or other catastrophic events
Azure provides multiple availability zones within a region. Each availability zone has one or more datacenters with redundancy built in. It is the responsibility of the services to provide AZ and regional resiliency
- **Network connectivity issues:** Loss of network connectivity between the client and the server, or within the Azure data centers

Azure provides multiple paths to connect source/target and automatically re-routes the traffic

- **DNS Failures:** Issues with DNS resolution that prevent access to the database. Please refer to [Resiliency in Azure DNS Private Resolver | Microsoft Learn](#) and the SLA provided <https://www.microsoft.com/licensing/docs/view/Service-Level-Agreements-SLA-for-Online-Services?lang=1>

Azure Service Outages

In this context, key services we are interested in are EntraID and Azure Key Vault as described below

- **Entra ID:** Flexible Server integrates this cloud-based identity and access management service. A user/role can be authenticated by Entra ID and access the Flexible Server. *Azure provides guidance on deploying resilient Entra ID service. Please refer to [Entra-id documentation](#)*
- **Azure Key Vault (AKV):** It is used to store customer managed keys for encryption. *AKV service is resilient to both availability zone and regional failures.*

Resilient Deployment of Flexible Server

Azure Database for Postgres flexible Server is designed to protect data and minimize downtime for mission-critical databases during both planned and unplanned events. Built on the resilient Azure infrastructure, flexible servers include features that ensure high availability and fault tolerance. These features are essential for maintaining business continuity and include fault protection mechanisms, rapid recovery capabilities, and measures to minimize data loss.

When architecting applications, it is crucial to consider the following objectives to ensure business continuity:

- **Recovery Time Objective (RTO):** This refers to the maximum acceptable amount of time an application can be offline. Different applications have varying tolerance levels for downtime; for example, a business-critical database demands much stricter uptime compared to a test database.
- **Recovery Point Objective (RPO):** This indicates the maximum acceptable amount of data loss measured in time. It is vital to assess how much data loss your business can tolerate in case of a disruption. Business critical applications require RPO=0 (i.e. no data loss)

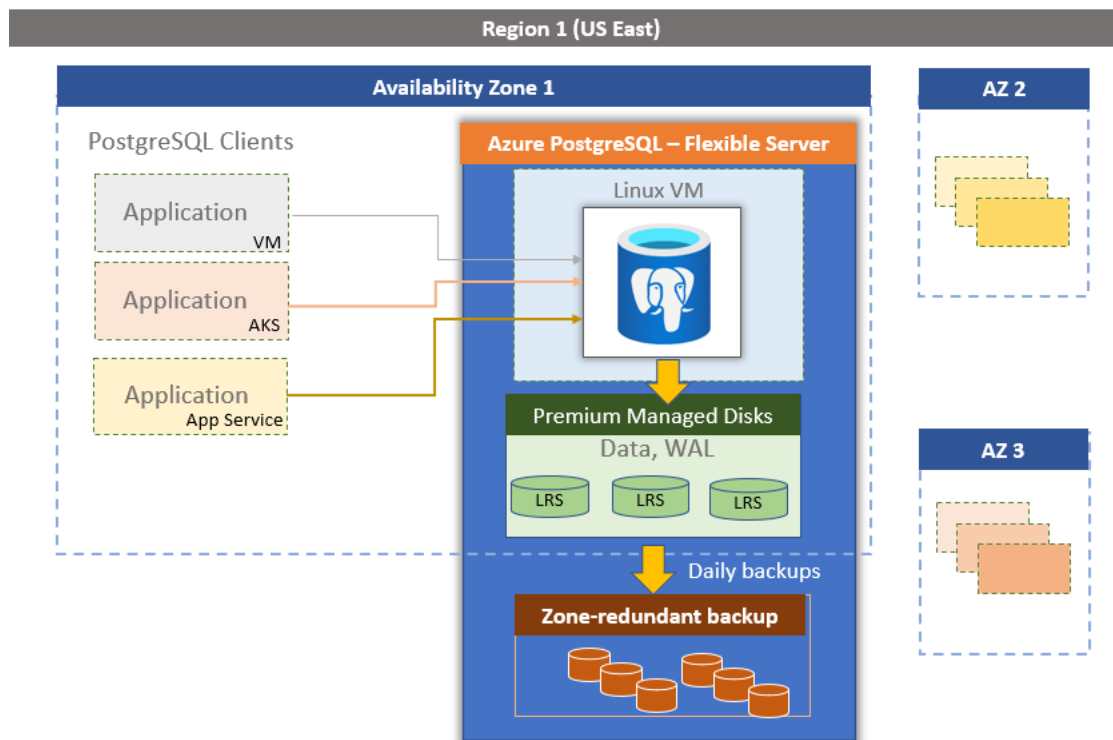
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The white paper describes different resiliency configurations in Azure Database for PostgreSQL flexible server. Each of these options offers a different RPO/RTO and the price point. It then recommends configurations for that can be ideal for your mission critical workloads.

Resiliency with Standalone Flexible Server

Configuration: Deploy a stand-alone Flexible Server in an availability zone (AZ) of your choice without enabling geo-backup as shown in the picture below.

Details: The Flexible Server is deployed in its own VM. The data and write-ahead log (WAL) are stored on a local redundant storage (LRS), premium managed disk (PMD) with three redundant copies in the same availability zone as the VM hosting Flexible Server. This allows the Flexible Server to access the storage with lower latency compared to if the storage was in a different AZ within the region. LRS storage is local to an availability zone and will become inaccessible if AZ goes down. In other words, LRS storage is not zone resilient. Unlike Data/WAL, the Flexible Server automatically archives the inactive portion of WAL on a zone-redundant storage (ZRS). Similarly, nightly backups are also stored on the ZRS storage. It is important to note that if AZ1, as shown in the picture below, the archived WAL and database backups are still accessible and can be used to create a new Flexible Server with point-in-time restore but at the cost of some data loss because archived WAL may not have all the committed transactions.



Resiliencies

Here are the resiliencies in this configuration

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- **VM (compute) resiliency:** If the compute node becomes unavailable, the Azure will automatically provision a new VM with a Flexible Server and bring up the service by running the recovery as data/WAL stored on LRS premium managed disks are available
- **Storage resiliency:** PMD (Premium managed disk) is an LRS(Local redundant storage) with three copies with automatic detection of disk fault (e.g. bit-rot) or disk failure. The disk write is done synchronously across the three disks therefore in case of disk issues, there is no data loss. Azure storage automatically detects disk failure and creates a new copy of the disk to maintain a quorum of three copies.
- **Zonal resiliency:** Weak AZ failure resiliency with potential of data loss (RPO > 0) and higher RTO
- **Regional resiliency:** None as there is no Geo Backup or cross region read replica is configured

Scenario	Resiliency	RPO	RTO
VM failure	Weak	0	Minutes
Storage failure	Strong	0	0
AZ failure	Weak	Minutes	Minutes
Region failure	None	NA	NA

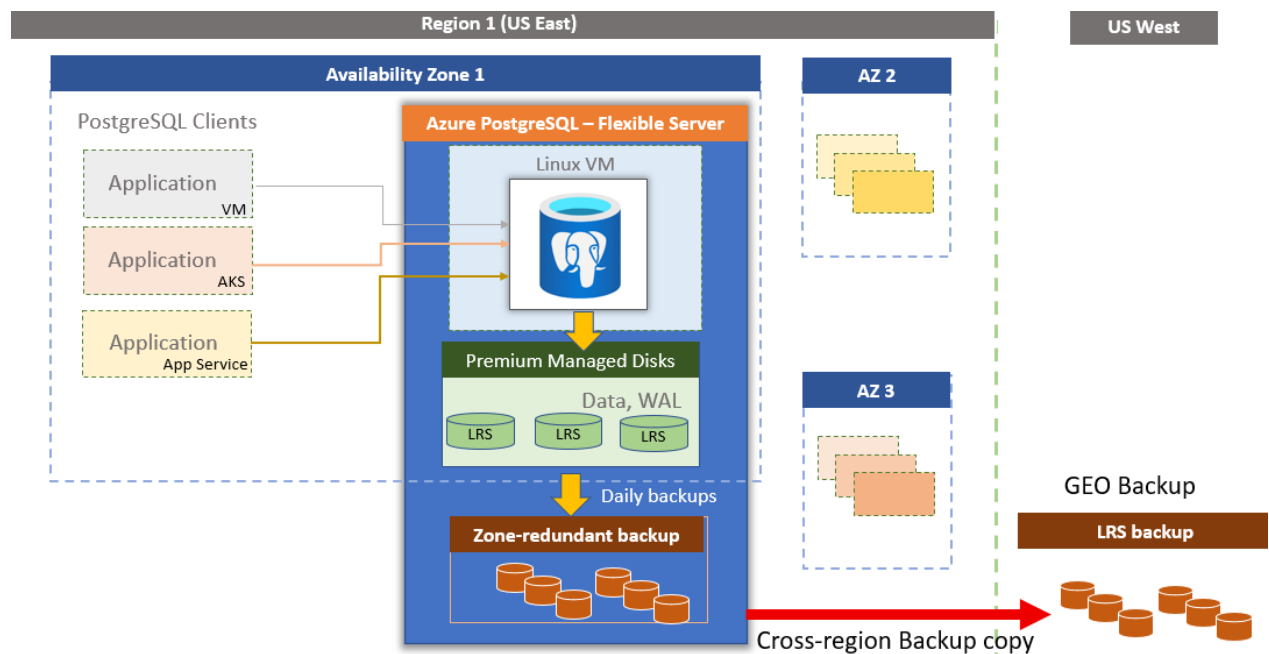
This configuration offers an up-time SLA of 99.9%

Resiliency with Geo Redundant Backup

Configuration: Deploy Flexible Server without HA but with Geo redundant backup enabled in the geo-paired region as shown in the picture below

Details: Azure Database for PostgreSQL flexible server details are the same as described in the previous configuration, but the archived WAL and snapshot database backups are asynchronously copied to a geo-paired azure region periodically. In our example, the primary region in the East US and its geo-paired region is West US. Please refer to [geo redundant backup documentation](#) for details. The Geo redundant backup is typically behind the primary Flexible Server by an hour or so. In case the primary server region becomes unavailable, you can bring your production workload online by restoring the Geo-backup albeit with some data loss.

Azure Database for PostgreSQL Flexible Server Resiliency Architecture



Resiliencies

Here are the resiliencies in this configuration

- **VM (compute) resiliency:** If the compute node becomes unavailable, the Azure will automatically provision a new VM with a Flexible Server and bring up the service by running the recovery as data/WAL stored on LRS premium managed disks are available
- **Storage resiliency:** PMD is an LRS with three copies with automatic detection of disk fault (e.g. bit-rot) or disk failure. The disk write is done synchronously across the three disks therefore in case of disk issues, there is no data loss. Azure storage automatically detects disk failure and creates a new copy of the disk to maintain a quorum of three copies.
- **Zonal Resiliency:** Weak AZ failure resiliency with potential of data loss (RPO > 0) and higher RTO
- **Regional Resiliency:** You can bring up your Flexible Server in the geo-paired region by running the recovery on geo-backup.

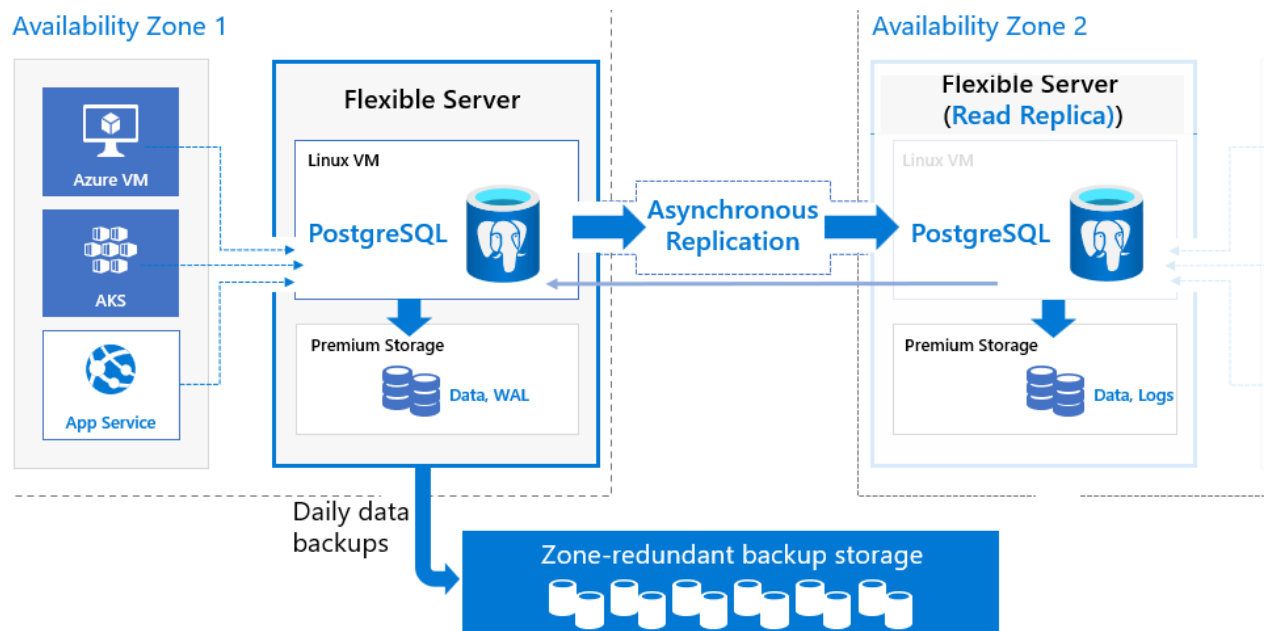
Scenario	Resiliency	RPO	RTO
VM failure	Weak	0	Minutes
Storage failure	Strong	0	0
AZ failure	Weak	Minutes or higher	Minutes or higher
Region failure	Weak	< 1 hour	Varies

This configuration offers an up-time SLA of 99.9%

Resiliency with Regional Read Replica

Configuration: Deploy Flexible Server without HA and [provision](#) a read replica in the same region as primary but in a different availability zone as shown in the picture below. For this configuration, we have not chosen Geo-Backup therefore no geo-resiliency.

Details: Flexible Server details are the same as described in the previous configuration except that a read replica is added in a different availability zone (AZ2). Any data changes, both DDL and DML, are sent to the read replica asynchronously using WAL. Primary Flexible Server does not wait for WAL to persist on the read replica before committing the transaction. This can lead to some data loss if you promote the read replica to primary. This configuration offers better zone resiliency with much lower RPO and RTO. Additionally, you can offload your read workload, typically reporting workload, to the read replica. The reads on the replica will have some latency, for example, at the peak transaction workload on the primary

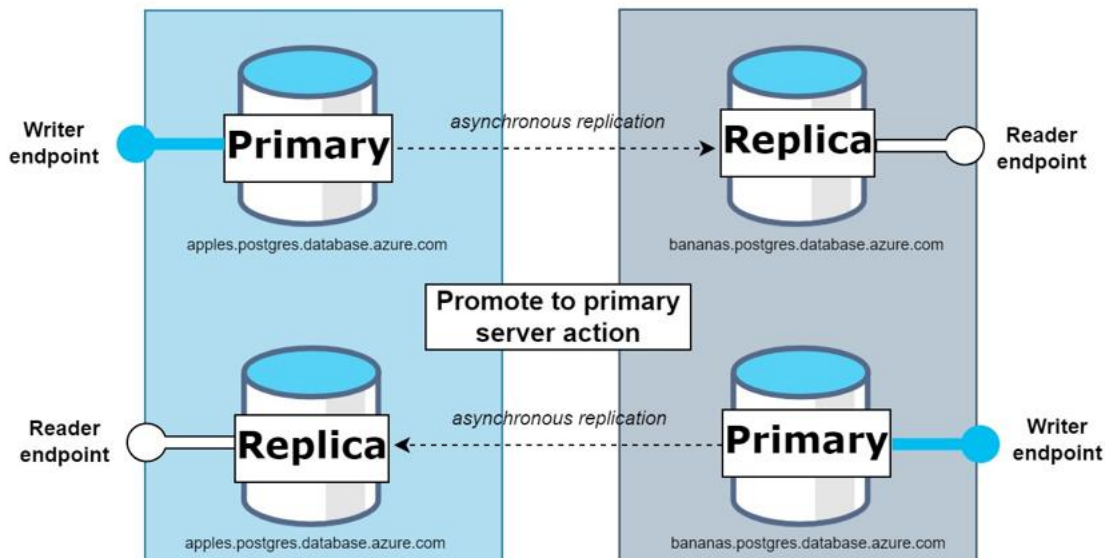


For AZ resiliency, there are two considerations. First, how do you detect that the primary AZ is down. Second, once you promote the read replica to primary, do I need to change the connection string. For detecting AZ failure, you will need to subscribe to Azure events that notify of the AZ failures and then manually trigger the promotion of read replica to primary. A common question asked is why we can't automatically fail over, and the primary reason is that there can be potential data loss. Customer must decide if want to wait for primary AZ to become healthy again or they would rather promote the read replica to the primary.

On the connection string question, the Flexible Server has simplified the transition with the introduction of [writer/reader endpoints](#). Your primary workload connects using the 'Writer'

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endpoint while the read workloads connect using the 'Read-only' endpoint which directs them to the read replica. In case of AZ failures, once you force promote the replica to be the primary, the 'Writer' endpoint as well as 'Read-only' endpoints will automatically point to the promoted read replica and therefore no connection string changes are required. Once the affected AZ comes online, the original primary will automatically resume, but now in a replica role.



Resiliencies

Here are the resiliencies in this configuration

- **VM (compute) resiliency:** If the compute node becomes unavailable, the Azure will automatically provision a new VM with a Flexible Server and bring up the service by running the recovery as data/WAL stored on LRS premium managed disks are available
- **Storage resiliency:** PMD is an LRS with three copies with automatic detection of disk fault (e.g. bit-rot) or disk failure. The disk write is done synchronously across the three disks therefore in case of disk issues, there is no data loss. Azure storage automatically detects disk failure and creates a new copy of the disk to maintain a quorum of three copies.
- **Zonal Resiliency:** Weak AZ failure resiliency with potential of data loss (RPO > 0) and higher RTO
- **Regional Resiliency:** None

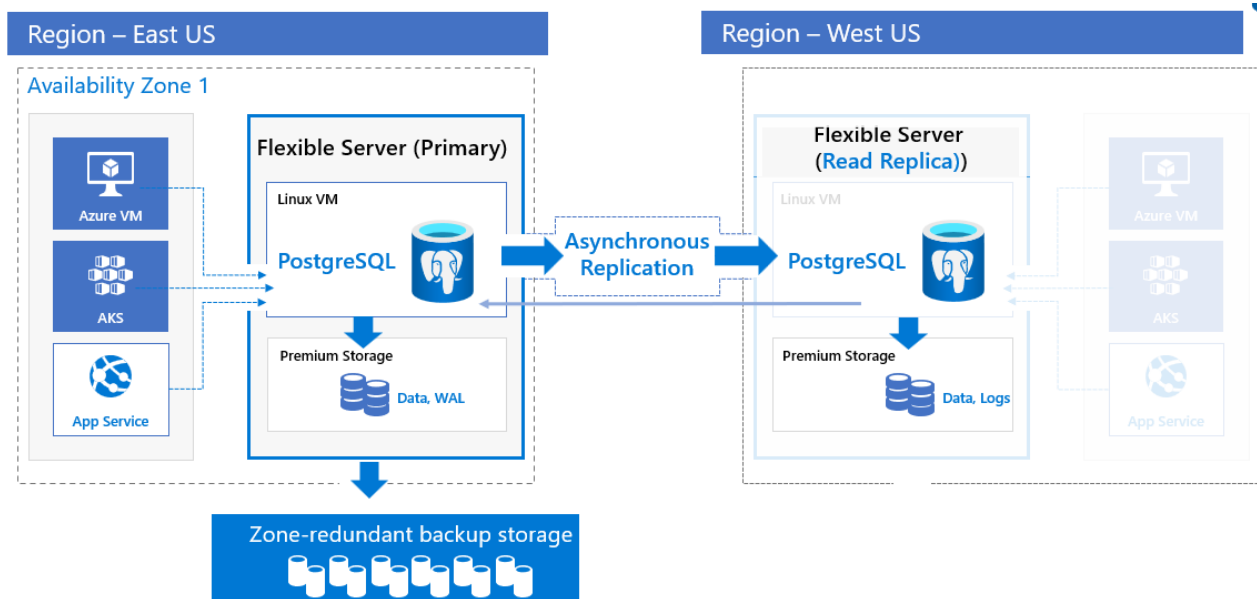
Scenario	Resiliency	RPO	RTO
VM failure	Weak	0	Minutes
Storage failure	Strong	0	0
AZ failure	Medium	Minutes or higher	Minutes of higher
Region failure	No	NA	NA

This configuration offers an up-time SLA of 99.9%

Geo-Resiliency with Read Replica

Configuration: Deploy Flexible Server without HA and provision a read replica in the geo-paired region as shown in the picture below.

Details: Flexible Server details are the same as described in the previous configuration except that a read replica is added in the geo-paired region. The geo-read replicas is a robust solution to achieve geo-disaster resiliency with much lower RTO/RPO compared to Geo-backup configuration. Additionally, customers can leverage geo-read replicas for queries. Like regional read-replica, you can use virtual endpoints to maintain the same connection string for the 'writer workload' in case primary region becomes unavailable. AZ resiliency is achieved thru zone resilient



Resiliencies

Here are the resiliencies in this configuration

- **VM (compute) resiliency:** If the compute node becomes unavailable, the Azure will automatically provision a new VM with a Flexible Server and bring up the service by running the recovery as data/WAL stored on LRS premium managed disks are available
- **Storage resiliency:** PMD is an LRS with three copies with automatic detection of disk fault (e.g. bit-rot) or disk failure. The disk write is done synchronously across the three disks therefore in case of disk issues, there is no data loss. Azure storage automatically detects disk failure and creates a new copy of the disk to maintain a quorum of three copies.

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- **Zonal Resiliency:** Weak AZ failure resiliency with potential of data loss (RPO > 0) and higher RTO
- **Regional Resiliency:** You can initiate a failover to geo-read replica and connect the production workload with the same connection string.

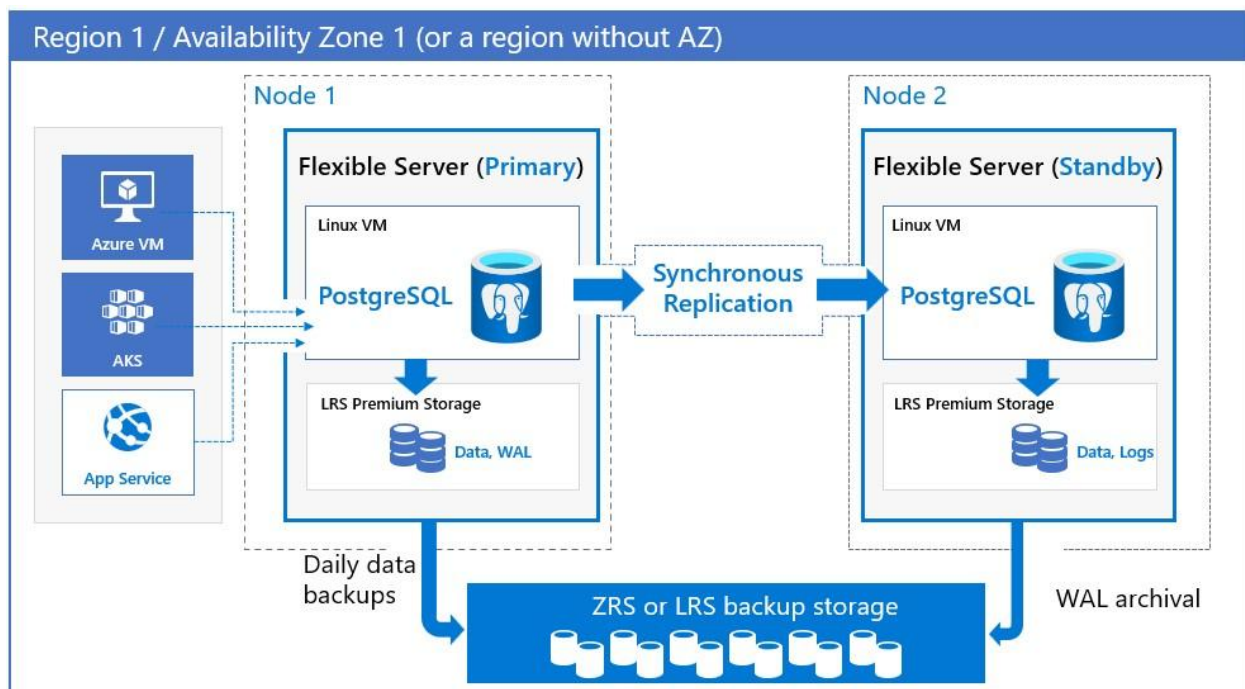
Scenario	Resiliency	RPO	RTO
VM failure	Weak	0	Minutes??
Storage failure	Strong	0	0
AZ failure	Weak	Minutes	Minutes
Region failure	Strong	< 5 minutes	Minutes

This configuration offers an up-time SLA of 99.9%

Resiliency with Zonal HA

Configuration: Deploy Flexible Server with HA but the standby is in the same AZ as the primary. For this configuration, we have not chosen Geo-Backup therefore no geo-resiliency as the focus is on the HA within the region

Details: Choose a zonal deployment when you want to achieve high availability within a single availability zone. Since both primary and standby are in the same AZ, the impact on the transactional latency with HA option is lower compared to if the standby was in a different AZ (discussed in the next section). With this configuration, you can lower the RTO should the primary node fail as standby will automatically get promoted to the primary with the same connection string.



Resiliencies

Here are the resiliencies in this configuration

- **VM (compute) resiliency:** If the primary compute node becomes unavailable, standby is automatically promoted to primary, and the application can continue to run with no connection string changes. It is important to know that during the failover, currently active transactions are aborted and need to be re-issued by the application. Since the primary/standby are connected with synchronous replication, there is no data loss (RPO=0) in case of failover.
- **Storage resiliency:** PMD is an LRS with three copies with automatic detection of disk fault (e.g. bit-rot) or disk failure. The disk write is done synchronously across the three disks therefore in case of disk issues, there is no data loss. Azure storage automatically detects disk failure and creates a new copy of the disk to maintain a quorum of three copies.
- **Zonal Resiliency:** Weak if the primary AZ fails because both primary and standby are in the same AZ. One AZ resiliency option is to restore using WAL archived and backup.
- **Regional Resiliency:** N/A

Scenario	Resiliency	RPO	RTO
VM failure	Strong	0	< 120 seconds
Storage failure	Strong	0	0
AZ failure	Weak	Minutes	minutes
Region failure	None	NA	NA

This configuration offers an up-time SLA of 99.9%

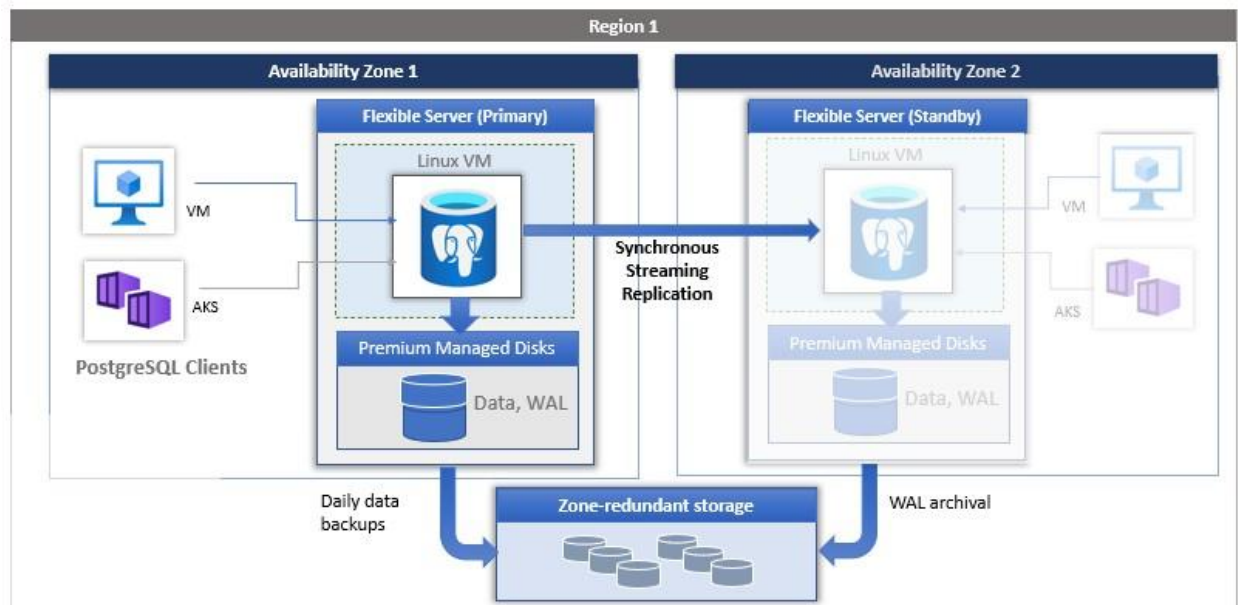
Zonal HA configuration offers uptime SLA of 99.95%

With Zone Redundant HA

Configuration: Deploy Flexible Server with HA but choose the standby in a different AZ. For this configuration, we have not chosen Geo-Backup therefore no geo-resiliency as the focus is on the HA within the region

Details: Choose a zone-redundant deployment when you want to achieve high availability within a region. This configuration offers AZ resiliency as the standby will automatically get promoted to the primary with the same connection string should the primary AZ fail with no data loss (i.e. RPO = 0). This configuration offers best uptime SLA of 99.99%

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Resiliencies

Here are the resiliencies in this configuration

- **VM (compute) resiliency:** If the primary compute node becomes unavailable, standby is automatically promoted to primary, and the application can continue to run with no connection string changes. It is important to know that during the failover, currently active transactions are aborted and need to be re-issued by the application. Since the primary/standby are connected with synchronous replication, there is no data loss (RPO=0) in case of failover.
- **Storage resiliency:** PMD is an LRS with three copies with automatic detection of disk fault (e.g. bit-rot) or disk failure. The disk write is done synchronously across the three disks therefore in case of disk issues, there is no data loss. Azure storage automatically detects disk failure and creates a new copy of the disk to maintain a quorum of three copies.
- **Zonal Resiliency:** Strong! Should the primary AZ fail, the standby is automatically promoted to primary in the other AZ and a new standby is provisioned in the remaining healthy AZ.
- **Regional Resiliency:** N/A

Scenario	Resiliency	RPO	RTO
VM failure	Strong	0	< 120 seconds
Storage failure	Strong	0	0
AZ failure	Strong	0	< 120 seconds
Region failure	None	NA	NA

This configuration offers an up-time SLA of 99.99%

Recommended Deployment Solutions

Now that we understand various resiliency vulnerabilities and how you can configure the Flexible Server to address those. Here are four recommended HA configurations for your workloads.

Key invariant in each of these solutions is zone-redundant HA, strongly recommended for any mission-critical workloads. Besides, each solution offers different geo-resiliency and the ability to offload read workload to read replicas. Though the focus of this paper is Flexible Server resiliency for HA configurations, it is important to have secure deployment of your mission critical workload by integrating with

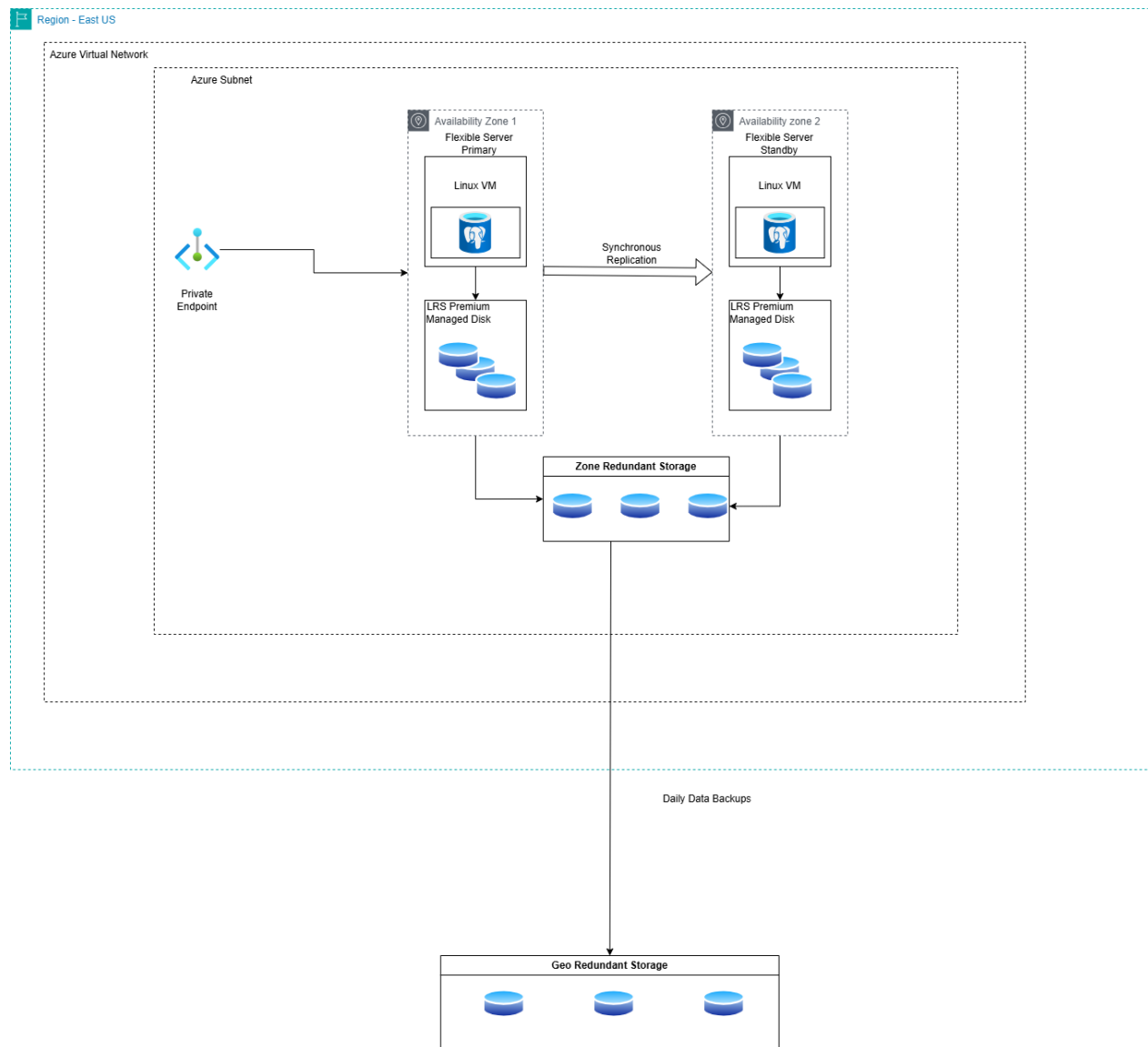
- EntraID for strong authentication and by following the golden rule by granting lowest privilege/permission to users/roles
- Private endpoint to protect/control your Server from external access
- Encryption with customer managed key (CMK) to have stronger control of the data and prevent any data leaks
- Use of virtual endpoints for configurations that have read replicas. These are read-write or read-only endpoints. When we perform promote to primary server action the endpoints automatically point to the correct instance after role change and you don't need to change the connection strings in the application

Zone Redundant HA with Geo-redundant backup

Deploying a standby instance in a different availability zone and enabling geo-redundant backups provides robust protection against both zonal and regional failures.

Zone-redundant HA provides resiliency for AZ failure and the Geo-redundant backups, configured during flexible server creation, ensure data is securely replicated to a paired region. This configuration provides a low-cost option for business continuity if the primary region becomes unavailable.

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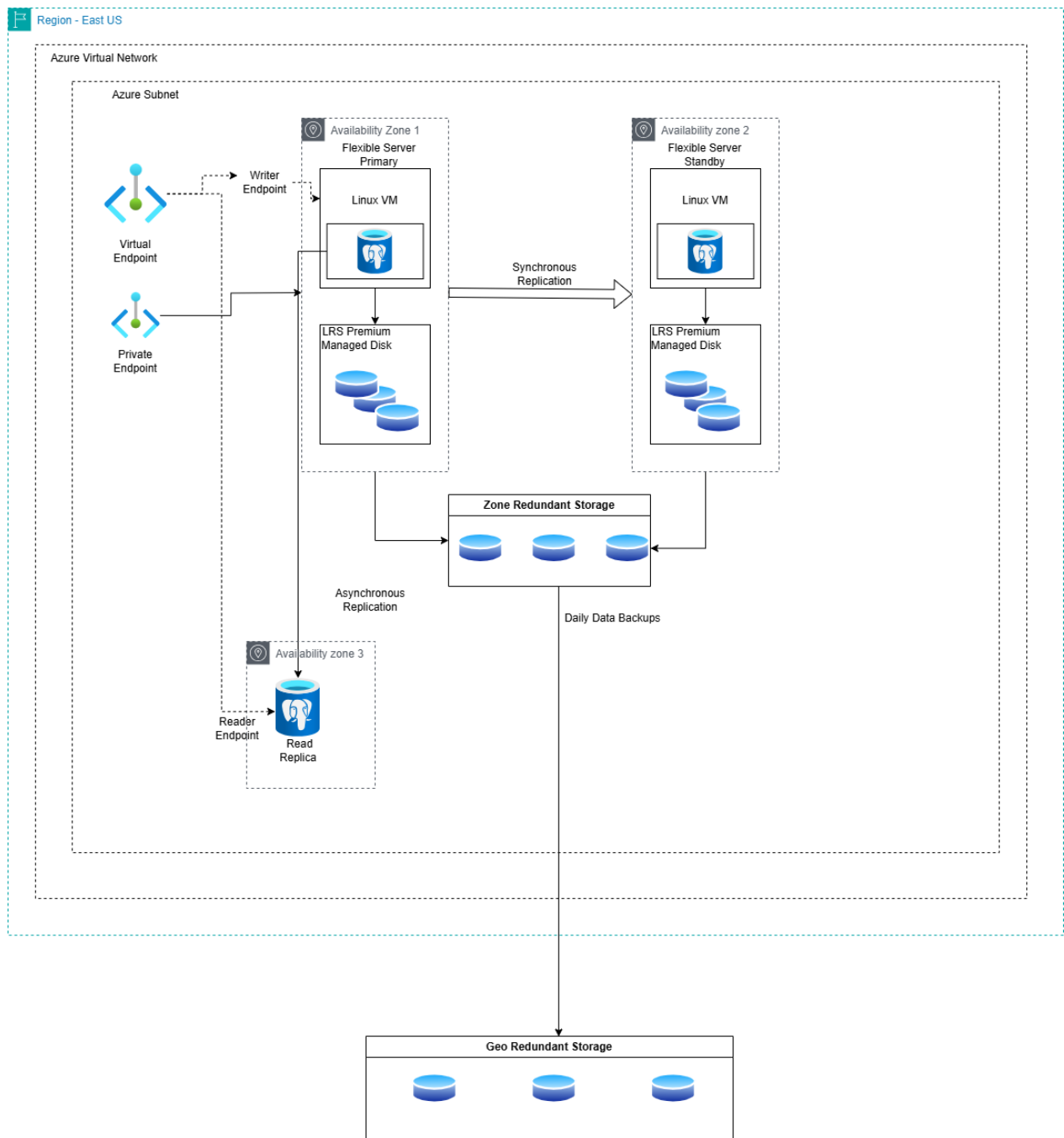


Scenario	Resiliency	RPO	RTO
VM failure	Strong	0	< 120 seconds
Storage failure	Strong	0	0
AZ failure	Strong	0	< 120 seconds
Region failure	Weak	< 1 hours	Varies

Zone Redundant HA with Read replica

This configuration deploys a flexible server instance with High Availability enabled using the "ZoneRedundant" option, geo-redundant backups, and a read replica in the same region as the primary. This setup provides protection against zonal outages and supports scaling out read-heavy workloads by directing read traffic to the replica through virtual endpoints. There is no impact on the resiliencies, but this configuration is recommended if you want to offload your read workload from the primary.

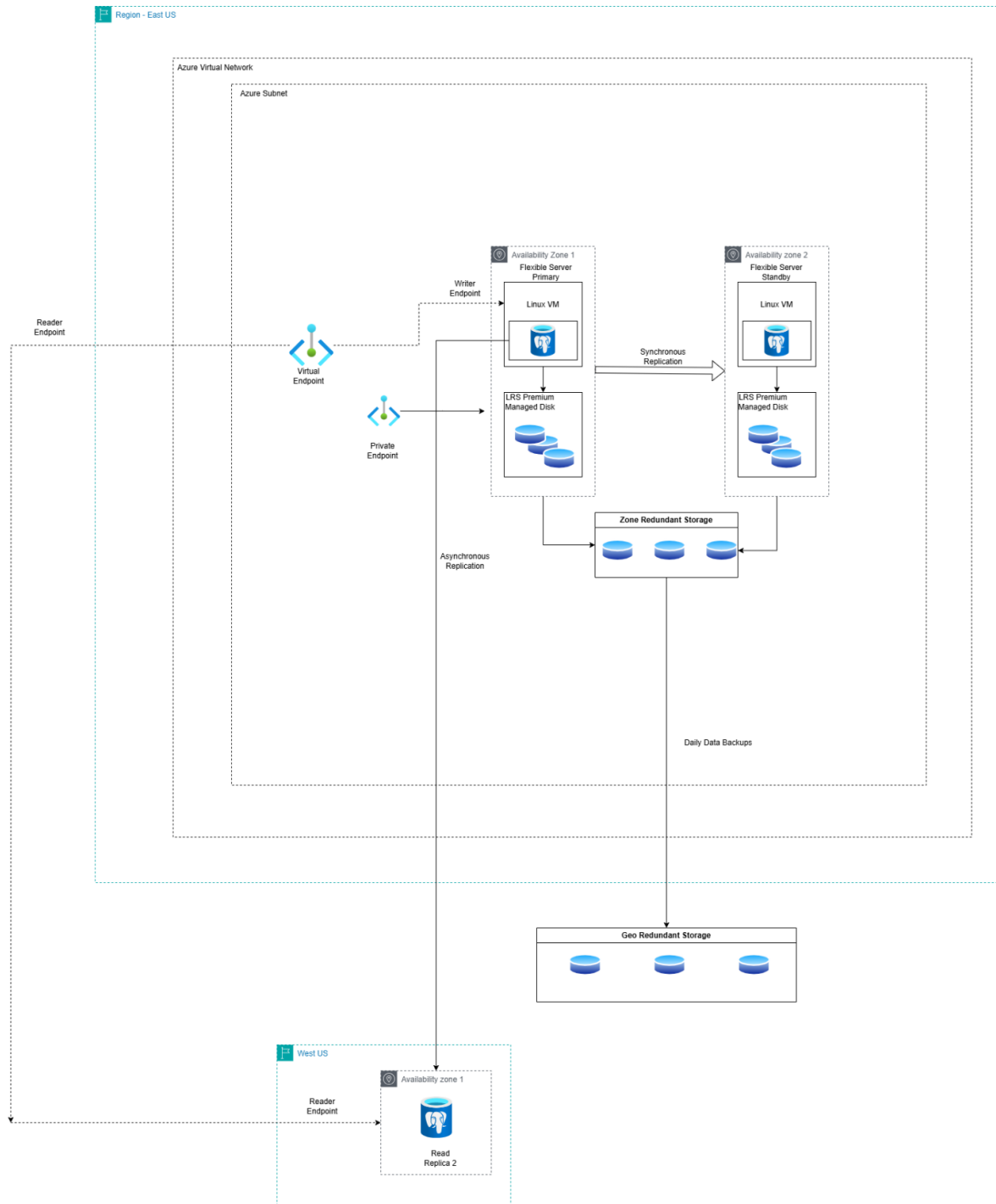
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Scenario	Resiliency	RPO	RTO
VM failure	Strong	0	< 120 seconds
Storage failure	Strong	0	0
AZ failure	Strong	0	< 120 seconds
Region failure	Weak	< 1 hours	Varies

Zone Redundant HA with Cross Region Read Replica

This configuration offers the best resiliency/price tradeoff. Zone-redundant HA offers zone resiliency while geo-read replica offers geo-resiliency. This configuration is best suited for workloads that either don't need to or cannot offload read workload to read-replica. Interestingly, customers can still offload read workload to the geo-read replica, but the latency will be higher compared to regional read-replica.

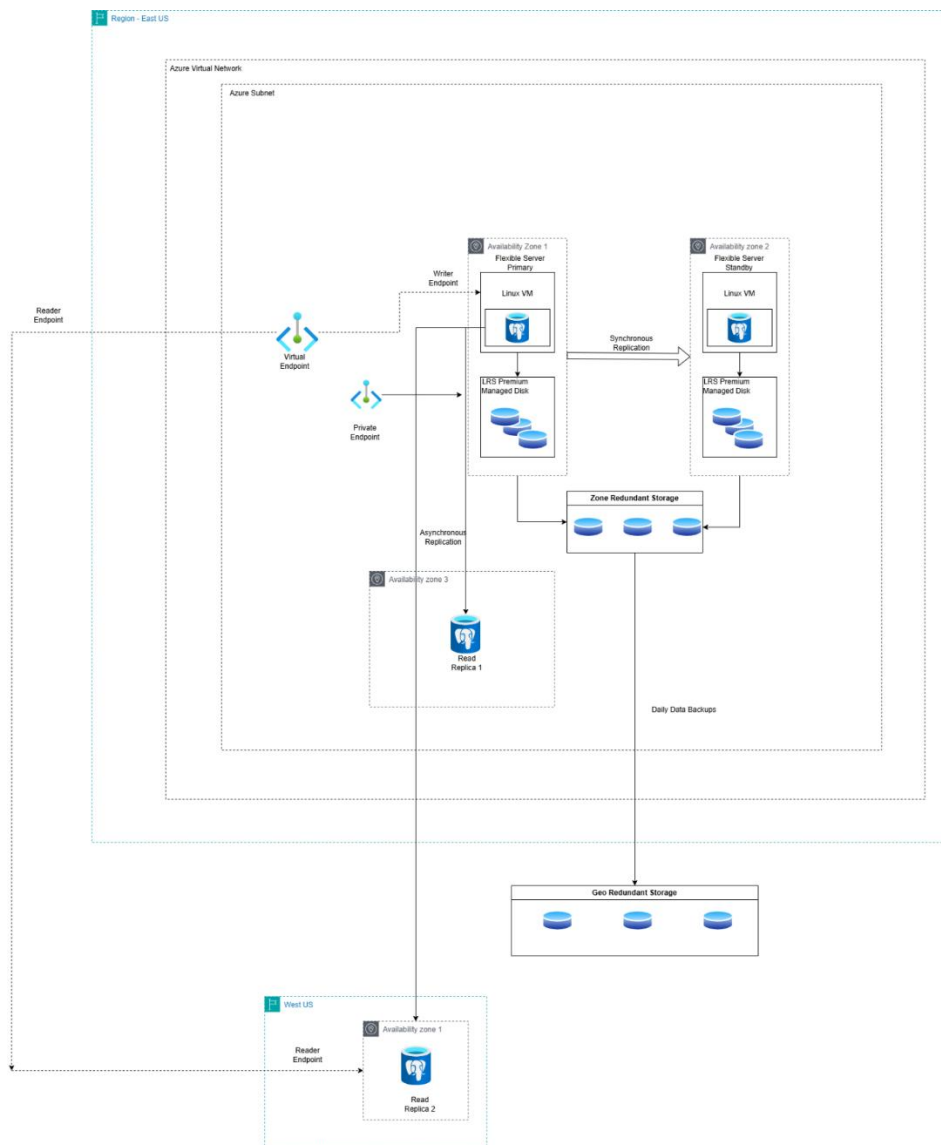


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Scenario	Resiliency	RPO	RTO
VM failure	Strong	0	< 120 seconds
Storage failure	Strong	0	0
AZ failure	Strong	0	< 120 seconds
Region failure	Strong	< 5 mins	Minutes

Zone Redundant HA with Regional and Cross-region Read Replica

This configuration offers the best resiliency/price tradeoff. Zone-redundant HA offers zone resiliency while geo-read replica offers geo-resiliency. This configuration is best suited for workloads can offload read workload to read-replica. Interestingly, customers can still offload read workload to the geo-read replica, but the latency will be higher compared to regional read-replica.



Scenario	Resiliency	RPO	RTO
VM failure	Strong	0	< 120 seconds
Storage failure	Strong	0	0
AZ failure	Strong	0	< 120 seconds
Region failure	Strong	< 5 mins	Minutes

Conclusion

Azure Database for PostgreSQL flexible server includes features designed to protect against regional and zonal failures. These features include High-Availability, geo-redundant backups, read-replicas, private link, virtual endpoints, and recommended architectures to enhance reliability and maximize uptime for mission-critical workloads. The architecture that provides the best RTO (Recovery Time Objective) and RPO (Recovery Point Objective) across all factors is Zone redundant HA with cross-regional read replica, which offers the highest resiliency among available architectures.

References:

- [Security overview](#)
- [Data encryption overview](#)
- [Entra Id overview](#)
- [Private Link Overview](#)
- [Data security and encryption best practices](#)
- [Virtual Endpoints](#)
- [Solution Accelerator](#)